Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to drive strategic business decisions

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Dataset Overview

3,900

Purchase Records

Transactional data analyzed

18

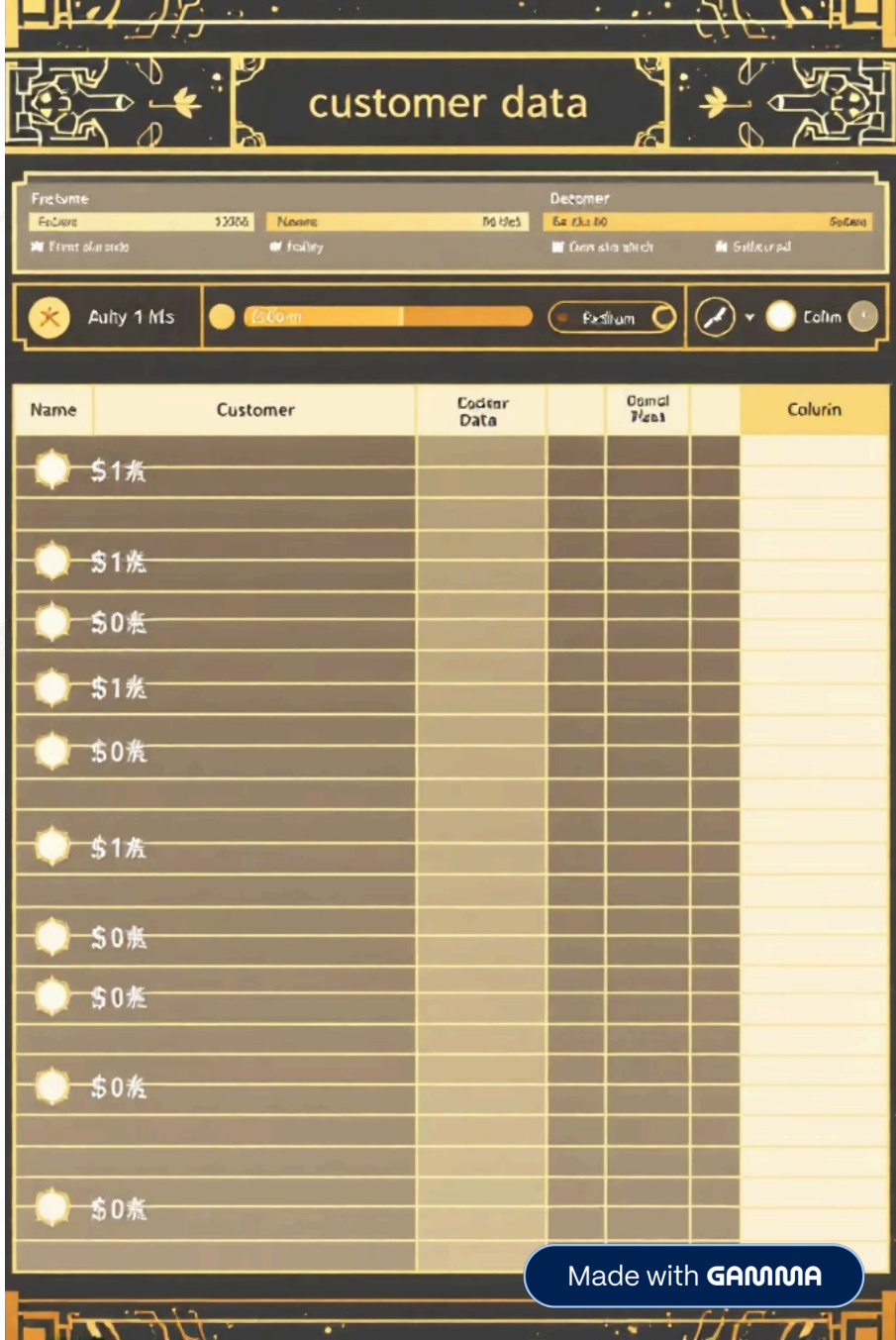
Data Fields

Customer demographics, purchase details, shopping behavior

37

Missing Values

Review ratings imputed



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Python Data Preparation

Data Cleaning

- Imported with pandas
- Checked structure and statistics
- Handled missing values
- Standardized column names

Feature Engineering

- Created age groups
- Calculated purchase frequency
- Dropped redundant columns
- Loaded to PostgreSQL

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SQL Analysis: Revenue Insights

Gender Revenue

Male: $157,890

Female: $75,191

Top Products

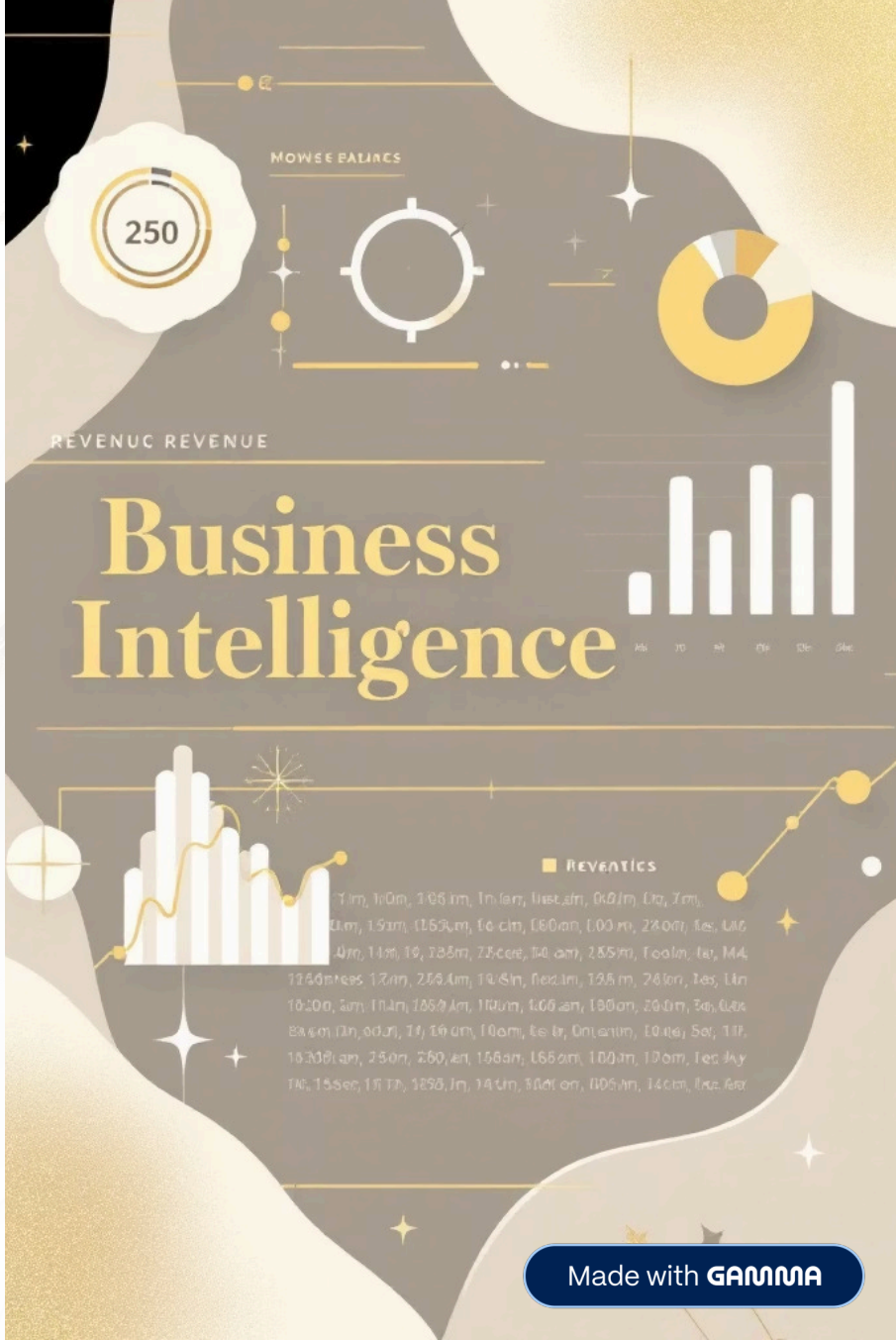
Gloves: 3.86 avg rating

Sandals: 3.84 avg rating

Shipping Impact

Standard: $58.46 avg

Express: $60.48 avg



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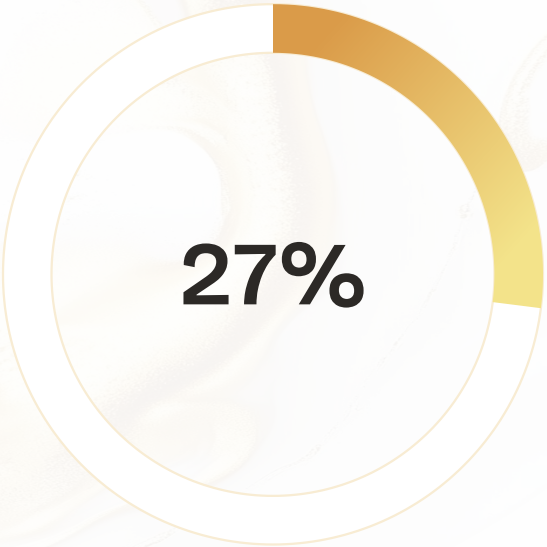
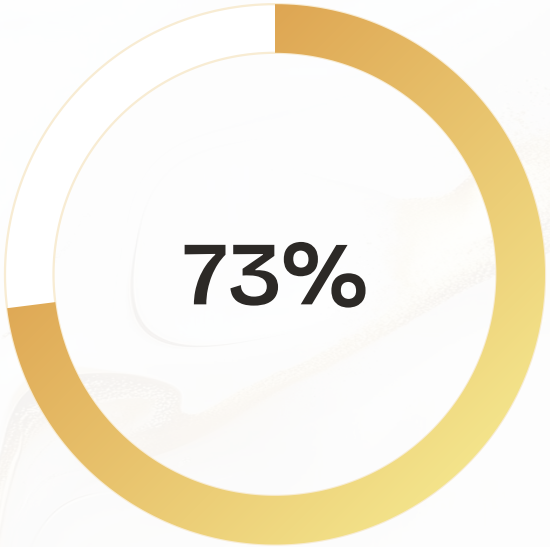
Subscription Analysis



Subscribers

1,053 customers



Non-Subscribers

2,847 customers

Average Spend

Subscribers: $59.49

Non-Subscribers: $59.87

Total Revenue

Subscribers: $62,645

Non-Subscribers: $170,436

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Product Performance

1

Discount-Dependent Products

Hats (50%), Sneakers (49.66%), Coats (49.07%)

2

Top Sellers by Category

Clothing: Blouse, Pants, Shirt

Accessories: Jewelry, Sunglasses, Belt

3

High-Spending Discount Users

839 customers used discounts above average purchase

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Customer Segmentation

Loyal

3,116 customers

High purchase frequency

Returning

701 customers

Multiple purchases

New

83 customers

First-time buyers

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Age Group Analysis

26%

Young Adult

$62,143 revenue, 1,000+ sales

24%

Middle-aged

$59,197 revenue, 900+ sales

22%

Adult

$55,978 revenue, 800+ sales

21%

Senior

$55,763 revenue, 700+ sales

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Power BI Dashboard

01

Interactive Visualizations

Revenue by category and age group

02

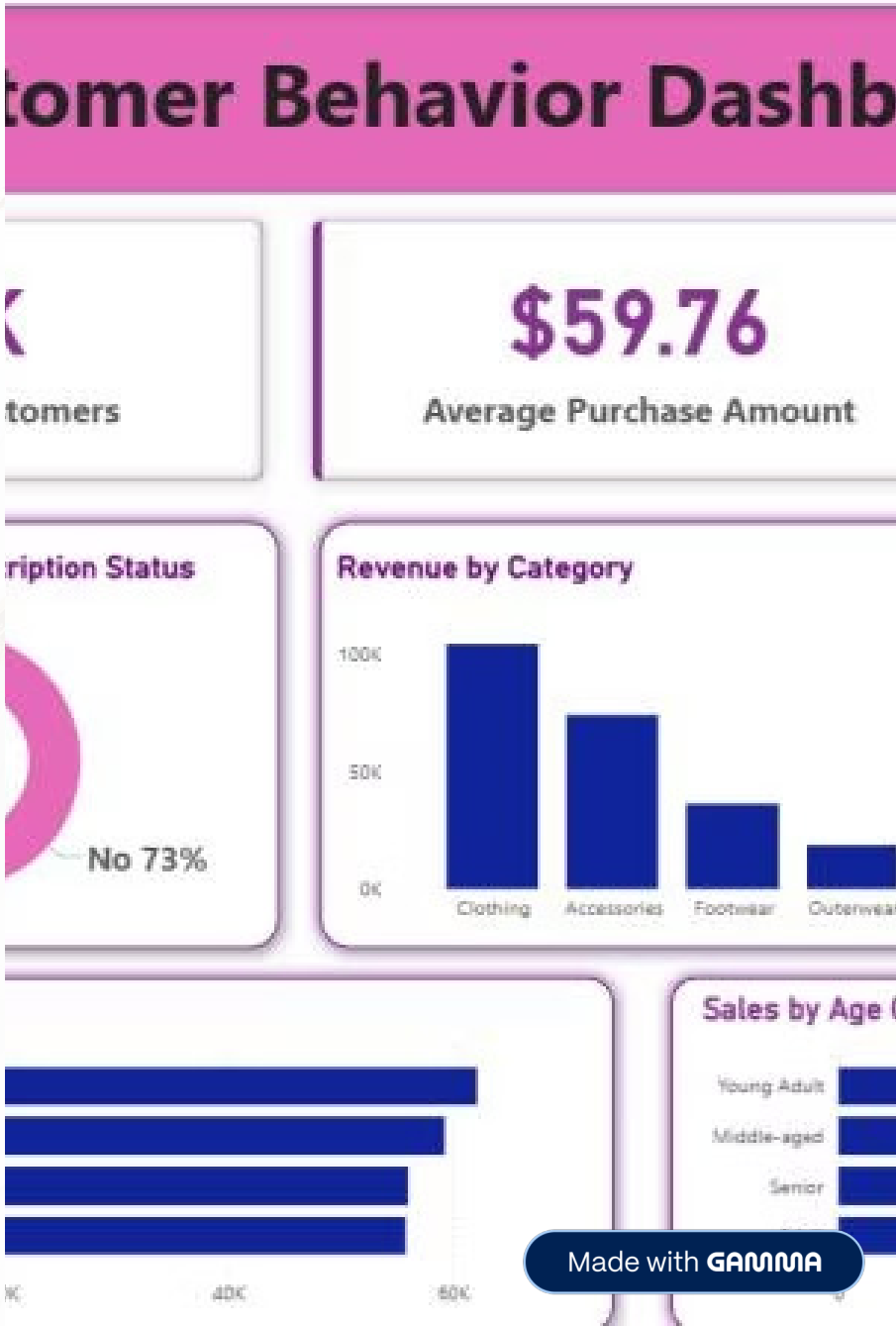
Key Metrics

3.9K customers, $59.76 avg purchase, 3.75 avg rating

03

Segment Analysis

Subscription status, gender, shipping type breakdowns



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Strategic Recommendations

1

Boost Subscriptions

Promote exclusive benefits

2

Customer Loyalty Programs

Reward repeat buyers

Review Discount Policy

Balance sales with margins

Targeted Marketing

Focus on high-revenue segments

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